# Negative & Directed Numbers

**Exam code:** 0607

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# Negative & Directed Numbers

## What are negative numbers?

- **Negative numbers** are any number **less than zero**
  - They may also be referred to as **directed** numbers
- Negative numbers are indicated by a minus sign (-)
  - To avoid confusion between subtraction and negative numbers, sometimes the following is used:
    - negative numbers are written in brackets
    - a longer dash is used for subtraction (—)
    - the minus for a negative number is raised (superscript), e.g. $^{-}4$
- Negative numbers are read by using the word 'negative' or 'minus' before the value
  - e.g. -8 would be read/said as "negative eight" or "minus eight"

## What are the rules for working with negative numbers?

- When **multiplying** and **dividing** with negative numbers
  - Two numbers with the **same sign** make a **positive**
    - $(-12) \div (-4) = 3$
    - $(-6) \times (-4) = 24$
  - Two numbers with **different signs** make a **negative**
    - $(-12) \div 4 = -3$
    - $6 \times (-4) = -24$
- When **adding** and **subtracting** with negative numbers
  - **Subtracting a negative** number is the same as **adding** the positive
    - e.g. $5 - (-3) = 5 + 3 = 8$
  - **Adding a negative** number is the same as **subtracting** the positive
    - e.g. $7 + (-3) = 7 - 3 = 4$

## Where are negative numbers used in real-life?

- **Temperature** is a common context for negative numbers

Help me understand

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Ask questions and get help with exactly what you're revising.
Got it

- If the temperature is -4°C, and it warms up by 6°C, the new temperature will be 2°C
  - This is equivalent to (-4) + 6 = 2
- To explain why (-5) - (-6) = 1, you could think of it as follows:
  - A room is -5°C, then -6°C of cold air is 'removed'
  - The room now warms to 1°C

- **Money** and **debt** is another common context for negative numbers
  - A negative sign means you owe money
  - If someone has a debt of $200, and they borrow another $400, their total debt is now $600
    - This is equivalent to (-200) + (-400) = -600
  - If someone is in debt by $300, but then pays off $200 of their debt, they are now only $100 in debt
    - This is equivalent to (-300) + 200 = -100

### Examiner Tips and Tricks

- Your calculator isn't always as clever as you may think!
- Using brackets around negative numbers will always make sure the calculator is doing what you want
  - e.g. The square of negative three is $(-3) \times (-3) = 9$
    On many calculators, $-3^2 = -9$ but $(-3)^2 = 9$
    The second one is the required calculation

### Worked Example

Complete the following table.

| Calculation | Answer |
|---|---|
| 3 + (-4) | |
| (-5) + (-8) | |
| 7 - (-10) | |
| (-8) - (-6) | |
| (-3) × 6 | |
| (-9) × (-2) | |
| (-9) ÷ (-3) | |
| (-10) ÷ 5 | |

**Answer:**

| Calculation | Working | Answer |
|---|---|---|

| | | |
|---|---|---|
| 7 - (-10) | 7 + 10 | 17 |
| (-8) - (-6) | (-8) + 6 | -2 |
| (-3) × 6 | 3 × 6 = 18<br>one is negative | -18 |
| (-9) × (-2) | 9 × 2 = 18<br>both are negative | 18 |
| (-9) ÷ (-3) | 9 ÷ 3 = 3<br>both are negative | 3 |
| (-10) ÷ 5 | 10 ÷ 5 = 2<br>one is negative | -2 |

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